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LAB 03	
Aim:	Implementation of input/output function using R
Theory:	**<u>Taking Input from User in R Programming:</u>** Developers often have a need to interact with users, either to get data or to provide some sort of result. Most programs today use a dialog box as a way of asking the user to provide some type of input. Like other programming languages in R, it's also possible to take input from the user. For doing so, there are two methods in R. ❖ **<u>cat()</u>** • Goal: Concatenating and printing items (such as text or numbers) to the terminal in an easy-to-use manner is the purpose of R's cat() function. Cat(), in contrast to print(), does not automatically insert a newline after each output or quote marks around characters. • Application: It can be used to print numerous things at once, and unless a different separator is given, it will automatically separate them with a space. When structuring output for reports or interactive sessions, it's especially helpful. ❖ menu() • Goal: An interactive menu where the user can choose an option is created in the console using R's menu() function. It is most helpful when creating basic user interfaces in the R console. It takes a vector of choices and displays them as a menu in the console. The user can choose an option by entering the corresponding number. It returns the selected option as an integer (the position of the choice). ❖ **<u>scan() and print()</u>** ❖ **<u>scan()</u>** • Purpose: The scan() function is used to read input from the console. It can be used to read data into R, either from the keyboard or from a file. It is particularly useful for reading raw data from the user interactively. • Usage: By default, scan() reads numeric data, but you can specify the type of data (e.g., character or logical). It allows users to input data one item at a time, and it returns the data as

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Code

❖ **3A**

third practical.R | third practical b.R | third practicle c.R

Source on Save | Run | Source

```
name<- readline(prompt = "enter your name :")
age <- as.numeric (readline(prompt="enter the age :"))
cat("hello",name,"! your are", age, "years old.\n")
```

❖ **3B**

third practical b.R | third practicle c.R | third practical.R

Source on Save

```
data <- data.frame(x1=c(1,2,2,3),
                   x2=c(4,5,6,7),
                   x3=c(8,9,10,11))
write.table(data,
            file="file.txt",
            row.names = FALSE)
data <- scan("file.txt",what = list("","",""))
print(data)
```

❖ **3C**

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```
choice <- menu(c("green","pink","orange"),title = "choose a color :")
cat("you choosed option :",choice,"\n")
```

Output:

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```
> name<- readline(prompt = "enter your name :")
enter your name :sheikh wasimuddin
> age <- as.numeric (readline(prompt="enter the age :"))
enter the age :19
> cat("hello",name,"! your are", age, "years old.\n")
hello sheikh wasimuddin ! your are 19 years old.
>
```

```
R 4.4.0 · ~/
> data <- data.frame(x1=c(1,2,2,3),
+                    x2=c(4,5,6,7),
+                    x3=c(8,9,10,11))
> write.table(data,
+             file="file.txt",
+             row.names = FALSE)
> data <- scan("file.txt",what = list("","",""))
Read 5 records
> print(data)
[[1]]
[1] "x1" "1"  "2"  "2"  "3"

[[2]]
[1] "x2" "4"  "5"  "6"  "7"

[[3]]
[1] "x3" "8"  "9"  "10" "11"

>
```

```
>
>
>
>
> choice <- menu(c("green","pink","orange"),title = "choose a color :")
choose a color :

1: green
2: pink
3: orange

Selection: 1
> cat("you choosed option :",choice,"\n")
you choosed option : 1
>
```

Conclusion	Data Manipulation Techniques executed successfully.
Name Of Faculty:	Prof. Akhil Jajulwar